

● MEDIUM

● ALGEBRA

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# Algebra

8 questions · ~12 min

# 19

Q1

ALGEBRA

For the linear function  $g$ , the graph of  $y = g(x)$  in the  $xy$ -plane has a slope of 2 and passes through the point 1, 14. Which equation defines  $g$ ?

- A.  $g(x) = 2x$
- B.  $g(x) = 2x + 2$
- C.  $g(x) = 2x + 12$
- D.  $g(x) = 2x + 14$

Q2

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Scientists collected fallen acorns that each housed a colony of the ant species *P. ohioensis* and analyzed each colony's structure. For any of these colonies, if the colony has  $x$  worker ants, the equation  $y = 0.67x + 2.6$ , where  $20 \leq x \leq 110$ , gives the predicted number of larvae,  $y$ , in the colony. If one of these colonies has 58 worker ants, which of the following is closest to the predicted number of larvae in the colony?

- A. 41
- B. 61
- C. 83
- D. 190

$$y = 18 - 5x$$

The equation above represents the speed  $y$ , in feet per second, of Sheila's bicycle  $x$  seconds after she applied the brakes at the end of a ride. If the equation is graphed in the  $xy$ -plane, which of the following is the best interpretation of the  $x$ -coordinate of the line's  $x$ -intercept in the context of the problem?

- A. The speed of Sheila's bicycle, in feet per second, before Sheila applied the brakes
- B. The number of feet per second the speed of Sheila's bicycle decreased each second after Sheila applied the brakes
- C. The number of seconds it took from the time Sheila began applying the brakes until the bicycle came to a complete stop
- D. The number of feet Sheila's bicycle traveled from the time she began applying the brakes until the bicycle came to a complete stop

Q4

GRID-IN

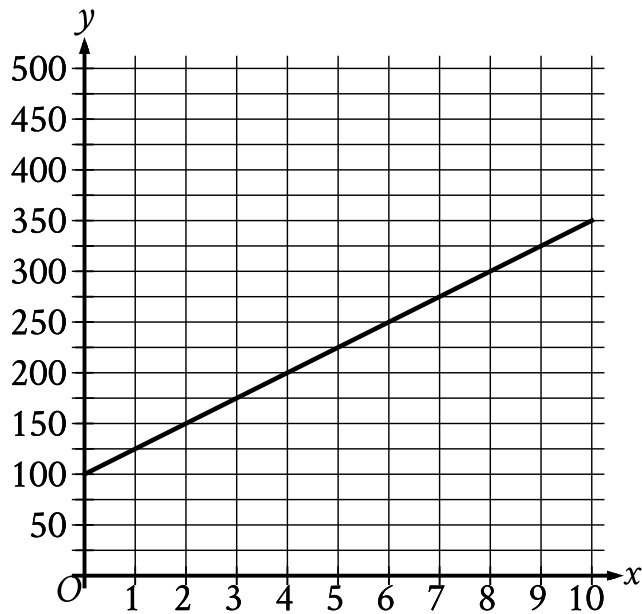
ALGEBRA

The function  $f$  is defined by  $fx = -9x + 9$ . What is the  $y$ -coordinate of the  $y$ -intercept of the graph of  $y = fx$  in the  $xy$ -plane?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



The graph of the function  $f$ , where  $y = f(x)$ , gives the total cost  $y$ , in dollars, for a certain video game system and  $x$  games. What is the best interpretation of the slope of the graph in this context?

- A. Each game costs \$25.
- B. The video game system costs \$100.
- C. The video game system costs \$25.
- D. Each game costs \$100.

| $x$ | $f(x)$ |
|-----|--------|
| 0   | -2     |
| 2   | 4      |
| 6   | 16     |

Some values of the linear function  $f$  are shown in the table above. What is the value of  $f(3)$ ?

- A. 6
- B. 7
- C. 8
- D. 9

$$f(x) = 39$$

For the given linear function  $f$ , which table gives three values of  $x$  and their corresponding values of  $fx$ ?

A.

| $x$ | $fx$ |
|-----|------|
| 0   | 0    |
| 1   | 0    |
| 2   | 0    |

B.

| $x$ | $fx$ |
|-----|------|
| 0   | 39   |
| 1   | 39   |
| 2   | 39   |

C.

| $x$ | $fx$ |
|-----|------|
| 0   | 0    |
| 1   | 39   |
| 2   | 78   |

D.

| $x$ | $fx$ |
|-----|------|
| 0   | 39   |
| 1   | 0    |
| 2   | -39  |

Q8

ALGEBRA

In the  $xy$ -plane, the graph of the linear function  $f$  contains the points 0, 3 and 7, 31.  
Which equation defines  $f$ , where  $y = fx$ ?

A.  $fx = 28x + 34$

B.  $fx = 3x + 38$

C.  $fx = 4x + 3$

D.  $fx = 7x + 3$